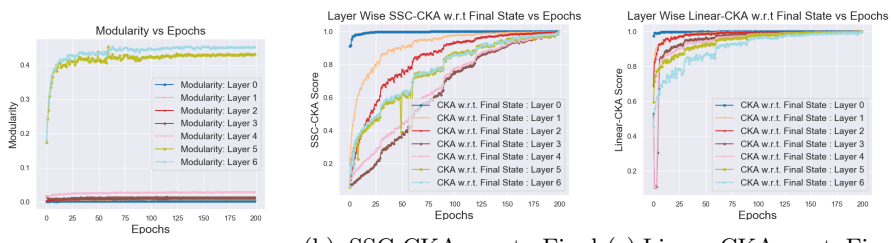
when compared to the deeper layers as shallower layers converge to their final representations much earlier in training when compared to deeper layers. This bottom-up convergence observation is similar to observations made in [15], and [20] as described earlier. This jump also signifies a marked deviation from the representations of the layer during it's previous state - before the step decay, and current state - after the step decay. These observations highlight the role of step decay in Stochastic Gradient Descent based training of Neural networks to escape saddle points and other spurious local minima.

4.3 Comparison with Linear-CKA:

As a comparison with related works in the area in Figure 1c and Figure 2c we present the same training dynamics as previous paragraph, this time by analyzing the Linear Kernel-CKA [10]. We observe a much cleaner bottom-up convergence of layers when compared with our method SSC-CKA, but the rates of convergence to final state is much faster when compared with the rates from SSC-CKA.

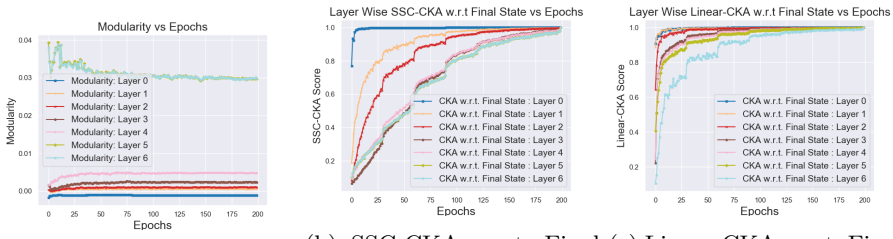


(a) Modularity vs Epochs (b) SSC-CKA w.r.t. Final State (c) Linear-CKA w.r.t. Final State

Fig. 1: Analysis of SSC Affinity Graphs using Modularity and CKA and it's comparison with a Linear Kernel Affinity Graph on DenseNet-121 network using CIFAR-10 dataset.

5 Analysis of Network Architecture

In this section, we utilise SSC-CKA to visualize network architectures. We do this by using CKA to compare the Subspace Affinity Graphs obtained by applying SSC on the activations of two different layers of the network. We try to focus on various inter-layer dynamics in different scenarios so as to learn intrinsic behaviours of the network and the architecture class it belongs to. In pursuit of these goals, we evaluate and demonstrate the architectural behaviour of various networks as a function of network architectural depth, network width, training duration in terms of epochs and finally as a function of quantity of training data.



(a) Modularity vs Epochs (b) SSC-CKA w.r.t. Final State (c) Linear-CKA w.r.t. Final State

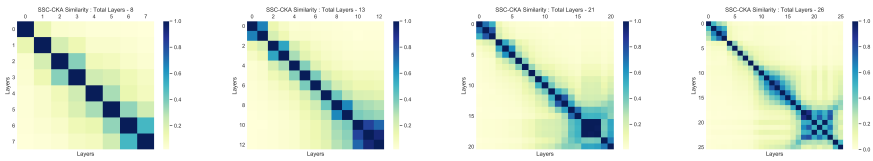
Fig. 2: Analysis of SSC Affinity Graphs using Modularity and CKA and it's comparison with a Linear Kernel Affinity Graph on DenseNet-121 network using CIFAR-100 dataset.

The motivation behind these experiments is to decipher how the latent space structure evolves through different layers of the network and how does a given set of layers add to the modeling power of the network. In doing so, we observe that deeper networks, wider networks, prematurely trained networks (networks trained for fewer epochs than optimal) and malnourished networks (networks trained on less training data but trained till saturation of performance) tend to develop prominent and mostly non-overlapping block diagonal structures in their layer-wise SSC-CKA heatmaps. This indicates that as the network's modeling capacity increases, a given amount training data is unable to exhaust the spare modeling bandwidth available due to additional layers, thus, a network tends to reorganise itself into blocks of layers with a high intra-block similarity among layers and a low inter-block similarity between layers. The networks used in these set of experiments are VGGs [22], ResNets [8], Wide-ResNets [24], all experiments conducted with CIFAR-10 [11] as the dataset.

5.1 Observing the Effects of Depth

In this section we demonstrate the effect of depth on neural network architecture by analyzing the SSC-CKA maps of different CNNs with varying depths. Through Figure 3a - Figure 3d, we demonstrate the effects of increasing depths in VGG architecture based networks. In Figure 3a, we train a VGG-11 and reach around 92% accuracy on the evaluation set. Applying SSC-CKA to this network, we observe a prominently diagonal similarity matrix indicating that most layers, especially the earlier ones learn unique representations and as we go deeper in the architecture, some inter layer similarity appears. Upon increasing the depth in the architecture to a VGG-16 as shown in Figure 3b, we observe a slight improvement in accuracy 94%, but we begin to observe a diagonally dominant similarity matrix where shallower layers have some degree of similarity with their neighbours, but deeper layers, especially towards the end tend to form blocks of layers which are very similar to each other indicating that the network doesn't

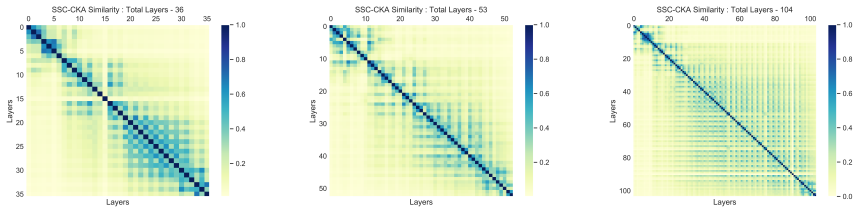
need to utilize the additional bandwidth to learn newer features in order to better learn and generalise. In Figure 3c, we show a similar observation for VGG-24 that attained an accuracy of 93%. However, as we increase the number of convolutional layers to 29 (VGG-29) Figure 3d, the performance of the network drops to around 62%, which is symptomatic of overfitting. We also observe a reinforcement in the prominence of the block diagonal structure signifying network over-parametrization relative to the data.



(a) VGG-11 (b) VGG-16 (c) VGG-24 (d) VGG-29

Fig. 3: Left to Right : Pairwise SSC-CKA between all convolution layers of VGG-11: (accuracy 92%), VGG-16: (accuracy 94%), VGG-24: (accuracy 93%) and VGG-29: (accuracy 62%) respectively on CIFAR-10 dataset
.

Next we demonstrate a similar set of results for ResNets of various depths, Figure 4a - Figure 4c. In case of ResNets, in addition to the block diagonal structure we also notice a chess-board pattern inside the blocks themselves, where every layer is similar to every alternate layer in a block. This is a consequence of skip-connections present inside resnet blocks, which allow inputs from one layer to propagate much deeper into the network. The observations in this experiment are in-line with the ones made in [10] and [18].



(a) ResNet-36 (b) ResNet-53 (c) ResNet-104

Fig. 4: Left to Right : Pairwise SSC-CKA between all convolution layers of ResNet-36: (accuracy 94%), ResNet-53: (accuracy 94%), ResNet-104: (accuracy 93%) respectively on CIFAR 10 dataset.